



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to GOWSALYA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

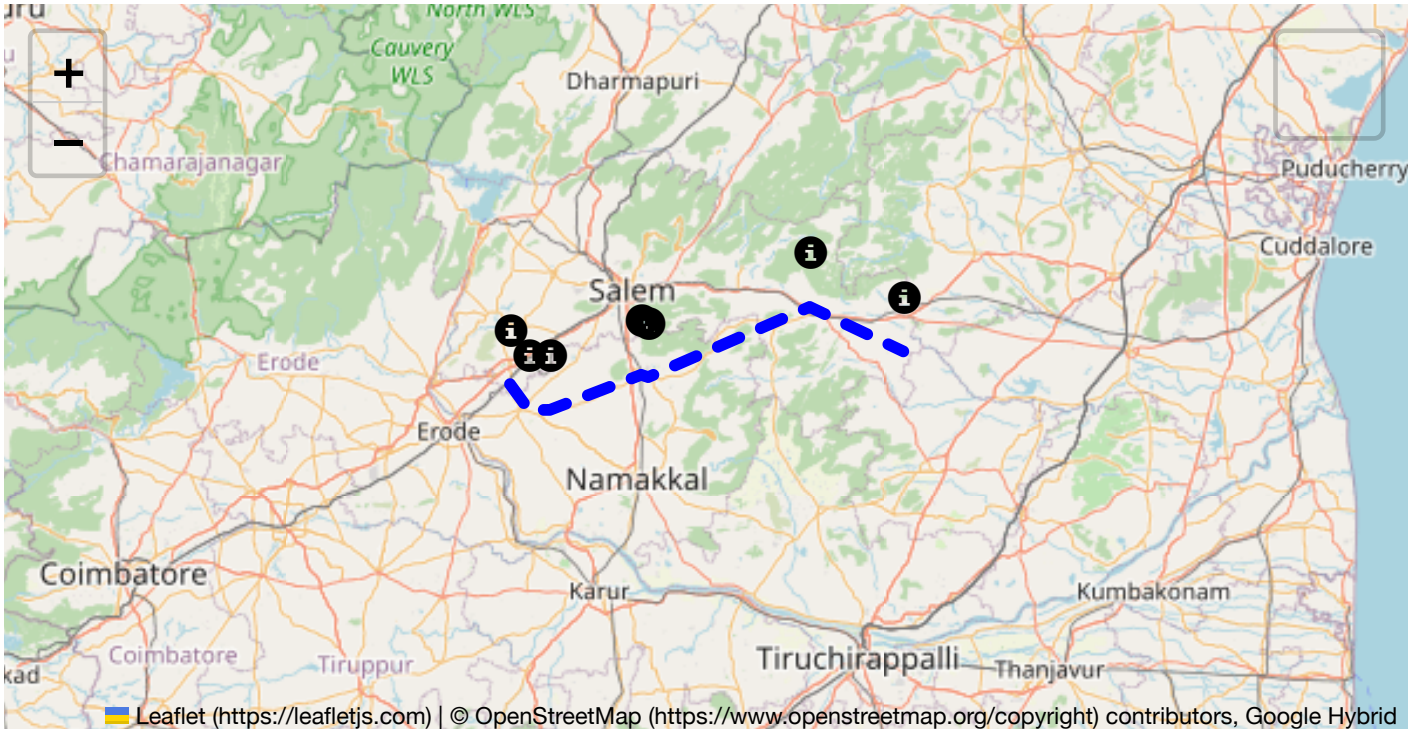
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

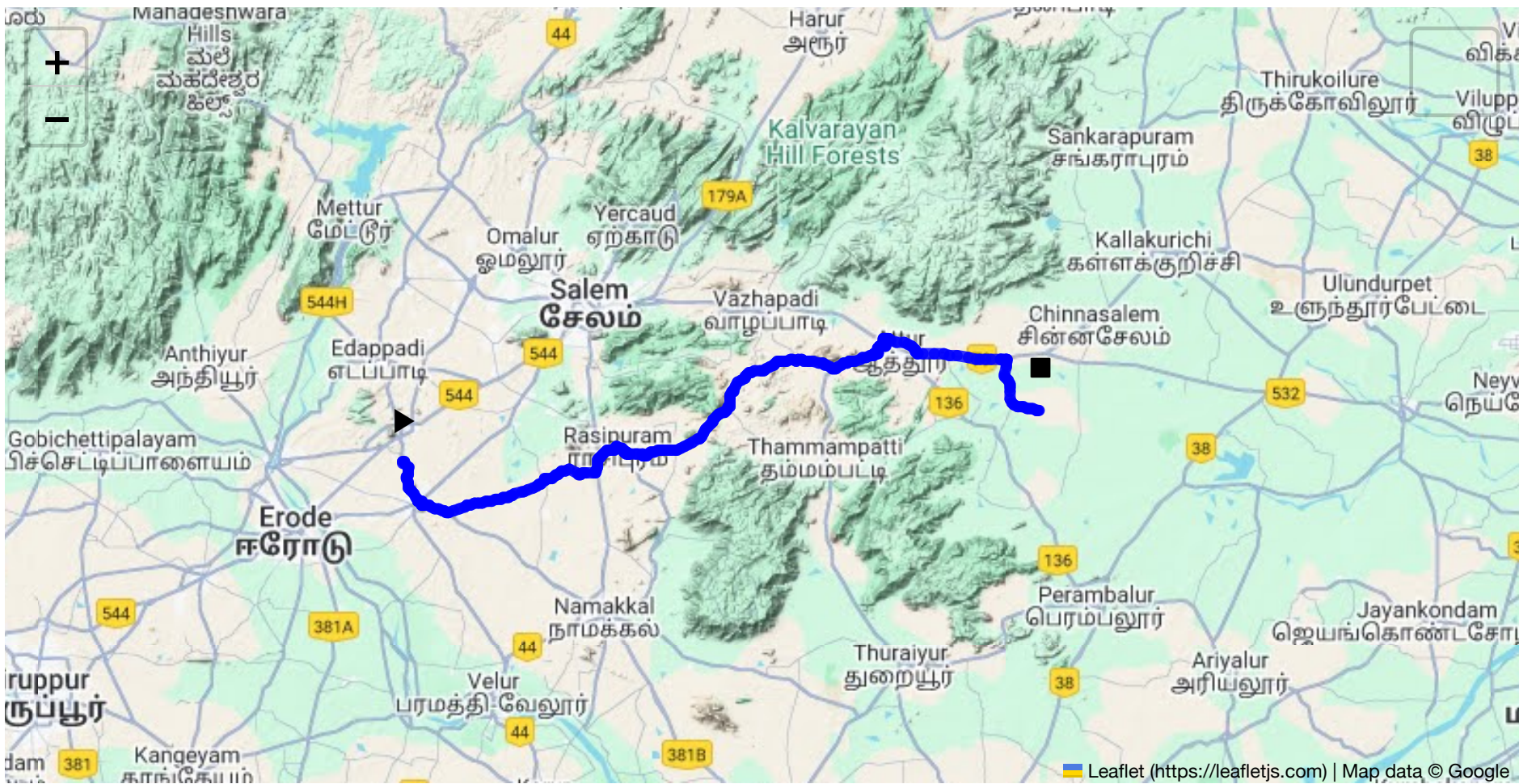
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 131.68 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours

Start: (11.4381, 77.8734)
End: (11.51324, 78.79597)



Welcome to the Journey Risk Management Study

- Overview of the Route Map:**
 - The route from CVQF+23W, Sangagiri, to GQ7W+79, Puliyankurichi, passes through key waypoints, including Namakkal, SH 79, Kavundampalaiyam, Rasipuram, and Koneripatti. The journey spans approximately 131.68 kilometers.
- Typical Weather Conditions and Potential Weather-Related Hazards:**
 - Tamil Nadu often experiences hot, humid weather. During monsoon (June to September), heavy rains can cause flooding and poor road visibility. Cyclonic activity, albeit rare, can affect the coastal regions and occasionally impact inland weather, causing heavy winds and rainfall.
- Analysis of Traffic Patterns:**
 - Peak traffic hours are typically from 8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM, especially in urban areas like Namakkal and Rasipuram. Congestion is common near city centers and markets, particularly during festive seasons.
- Assessment of Road Quality and Infrastructure:**
 - State highways like SH 79 generally maintain good conditions but can present potholes and uneven surfaces, particularly after the monsoon season. Some stretches may lack proper lighting, affecting night-time visibility.
- Suggestions for Alternative Routes for Emergencies:**
 - If an alternate route is necessary, consider using NH 44 for bigger sections, as it offers better facilities and connectivity. State highways SH 95 or SH 86 can serve as detours, depending on the specific emergency location.
- Summary of Local Regulations Affecting Hazardous Material Transport:**
 - Transporting hazardous materials requires special permits. Regulations mandate specific labeling and precautions for additional safety. There are designated routes for such materials,

often avoiding highly populated areas.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- Past incidents often involve driver fatigue or mechanical failure. While rare, mishaps could lead to road closures and require adherence to safety protocols and emergency management plans.

8. Environmental Considerations and Sensitive Areas:

- The route traverses agricultural zones and rural landscapes. Drivers should be aware of wildlife crossing areas and avoid spillage or littering to prevent environmental harm.

9. Analysis of Communication Coverage:

- Communication coverage is generally reliable in urban areas, though rural stretches might experience brief connectivity issues. Major providers cover the region, but it is advisable to have alternate means of communication.

10. Estimated Emergency Response Times:

- In urban areas, emergency response could take 15-30 minutes. In rural stretches or during peak traffic, response times might extend to one hour or more.

11. Overall Summary of Risk Assessment:

- The route presents challenges typical of regional travel in India, including variable road quality and congestion. Weather and emergency management are critical considerations due to potential for flooding and extended response times. Drivers should be well-versed in local regulations for hazardous materials and environmental protection measures. Proper route planning and adherence to safety protocols can minimize risks associated with this transit path.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	U-Turn	High	11.4535263, 78.1964113	10 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	Medium	11.38376, 77.89914	30 KM/Hr
6	Turn	High	11.38186, 77.90258	15 KM/Hr
7	Turn	High	11.37828, 77.90171	15 KM/Hr
8	Turn	Medium	11.36754, 77.93919	30 KM/Hr
9	Turn	Medium	11.41354, 78.08075	30 KM/Hr
10	Turn	High	11.41447, 78.08063	15 KM/Hr
11	Turn	High	11.42964, 78.11587	15 KM/Hr
12	Turn	Medium	11.42447, 78.15264	30 KM/Hr
13	Turn	High	11.42611, 78.15532	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	High	11.44521, 78.15833	15 KM/Hr
15	Turn	Medium	11.45566, 78.17361	30 KM/Hr
16	Turn	High	11.45587, 78.18057	15 KM/Hr
17	Blind Spot	Blind Spot	11.46035, 78.18534	10 KM/Hr
18	Turn	High	11.45929, 78.18544	15 KM/Hr
19	Turn	Medium	11.45764, 78.19208	30 KM/Hr
20	Blind Spot	Blind Spot	11.45084, 78.20378	10 KM/Hr
21	Blind Spot	Blind Spot	11.45354, 78.19642	10 KM/Hr
22	Turn	Medium	11.45117, 78.20024	30 KM/Hr
23	Turn	Medium	11.44993, 78.22152	30 KM/Hr
24	Turn	Medium	11.44896, 78.22294	30 KM/Hr
25	Turn	High	11.44797, 78.22314	15 KM/Hr
26	Turn	Medium	11.44756, 78.22414	30 KM/Hr
27	Turn	Medium	11.45353, 78.23584	30 KM/Hr
28	Turn	Medium	11.49835, 78.32417	30 KM/Hr
29	Turn	Medium	11.50784, 78.33331	30 KM/Hr
30	Turn	Medium	11.50865, 78.33966	30 KM/Hr
31	Turn	Medium	11.51183, 78.34416	30 KM/Hr
32	Turn	Medium	11.51870, 78.34816	30 KM/Hr
33	Turn	Medium	11.54842, 78.35627	30 KM/Hr
34	Turn	Medium	11.56719, 78.37813	30 KM/Hr
35	Turn	Medium	11.56940, 78.39856	30 KM/Hr
36	Turn	Medium	11.57497, 78.49389	30 KM/Hr
37	Turn	High	11.57093, 78.49993	15 KM/Hr
38	Turn	Medium	11.57829, 78.51245	30 KM/Hr
39	Turn	Medium	11.59108, 78.55637	30 KM/Hr
40	Turn	Medium	11.59253, 78.55778	30 KM/Hr
41	Turn	High	11.59333, 78.56209	15 KM/Hr
42	Turn	High	11.60406, 78.56494	15 KM/Hr
43	Turn	Medium	11.60471, 78.57359	30 KM/Hr
44	Turn	High	11.60491, 78.57377	15 KM/Hr
45	Turn	Medium	11.61294, 78.57255	30 KM/Hr
46	Turn	High	11.61403, 78.57272	15 KM/Hr
47	Turn	Medium	11.59279, 78.62141	30 KM/Hr
48	Turn	Medium	11.58562, 78.74231	30 KM/Hr
49	Turn	Medium	11.58530, 78.74272	30 KM/Hr
50	Turn	High	11.58553, 78.74952	15 KM/Hr
51	Turn	High	11.52182, 78.75791	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
52	Turn	Medium	11.51306, 78.79502	30 KM/Hr
53	Turn	High	11.51331, 78.79585	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
14	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
15	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
16	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
17	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
18	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
19	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
20	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
21	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium
22	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
23	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium
24	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium
26	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium
27	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
28	clinic	Dr. Saravanakumar Clinic	11.5842002, 78.4441223	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	hospital	Government Hospital	11.572472, 78.497407	30 km/h	Medium
31	hospital	Abi Hospital	11.59319, 78.560129	30 km/h	Medium
34	clinic	Sugam Clinic	11.5832481, 78.7496438	30 km/h	Medium
35	hospital	Selvam Hospital	11.585978, 78.748761	30 km/h	Medium
36	hospital	Prabushankar Hospital	11.584239, 78.7490323	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
10	school	SRV EXCEL Matriculation Higher Secondary School	11.4299679, 78.152616	30 km/h	Medium
11	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
12	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
13	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium
25	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium
30	school	Rasi Matric HR Sec School	11.5760027, 78.5062808	30 km/h	Medium
32	school	MARUTHI PRIMAR, SECONDAR ,HIGHER SECONDARY SHOOL	11.5859471, 78.7186322	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
33	college	Maruthi Polytechnic College	11.5856852, 78.7201072	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.4535263, 78.1964113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



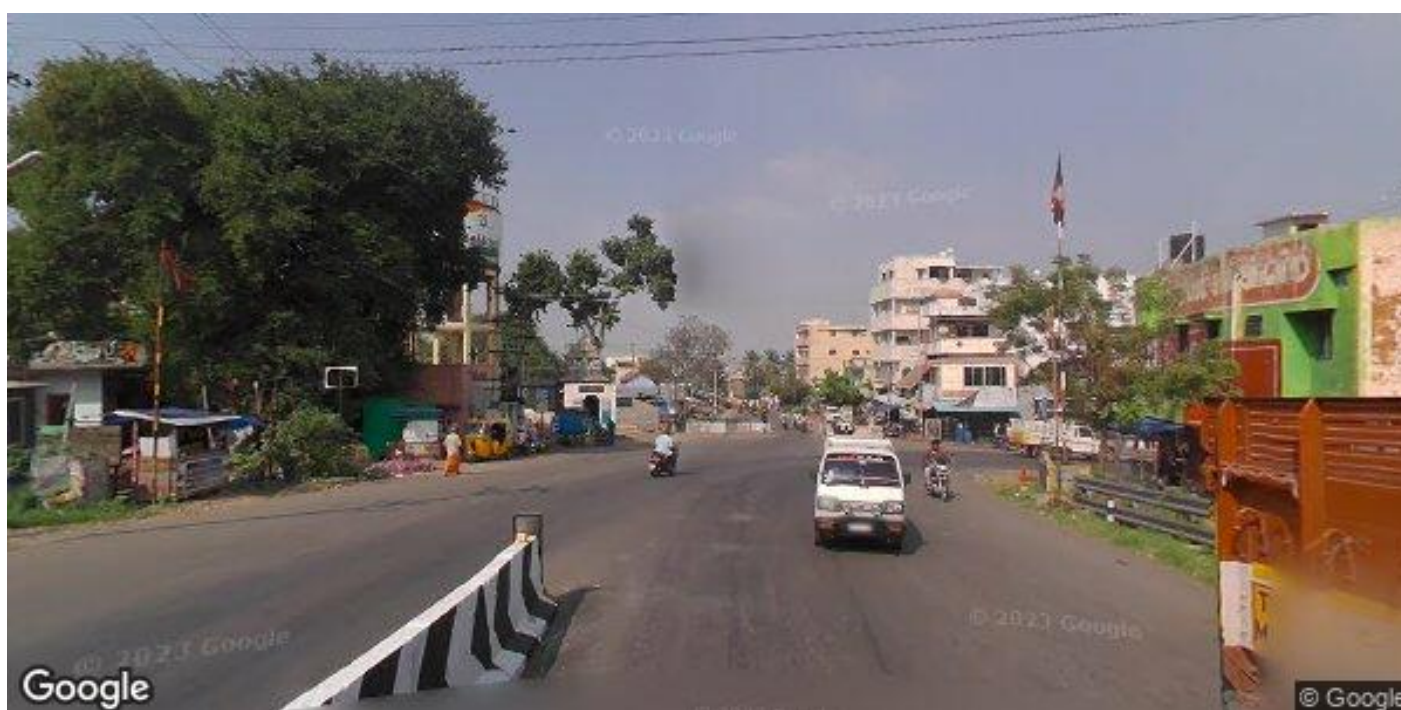
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38376, 77.89914



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37828, 77.90171



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36754, 77.93919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41354, 78.08075



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41447, 78.08063



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42447, 78.15264



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42611, 78.15532



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44521, 78.15833



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45566, 78.17361



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45587, 78.18057



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.46035, 78.18534



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45929, 78.18544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 78.19208



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45084, 78.20378



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45354, 78.19642



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45117, 78.20024



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44993, 78.22152



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44896, 78.22294



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44797, 78.22314



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44756, 78.22414



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 78.23584



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49835, 78.32417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50784, 78.33331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50865, 78.33966



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51183, 78.34416



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51870, 78.34816



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54842, 78.35627



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56719, 78.37813



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56940, 78.39856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57497, 78.49389



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57093, 78.49993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57829, 78.51245



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59333, 78.56209



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60406, 78.56494



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60471, 78.57359



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60491, 78.57377



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61294, 78.57255



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61403, 78.57272



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59279, 78.62141



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58562, 78.74231



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58530, 78.74272



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58553, 78.74952



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.52182, 78.75791

